Does ACOSQE outperform generative models on implicit opinion extraction?

ACOSQE adalah tugas, bukan arsitektur model pembanding

Consensus Meter

Requires at least 5 papers that directly answer your question. Try adjusting your query to find more papers.

Tinjauan literatur menunjukkan bahwa pertanyaan ini didasarkan pada perbandingan kategori yang keliru: ACOSQE adalah formulasi tugas, sedangkan model generatif merupakan salah satu keluarga metode untuk menyelesaikannya.

FIGURE 1 Konsensus literatur mengenai kategorisasi tugas ACOSQE vs pendekatan generatif

ACOSQE (*Aspect-Category-Opinion-Sentiment Quadruple Extraction*) merujuk pada tugas mendeteksi target aspek, kategori aspek, istilah opini, dan polaritas sentimen sekaligus, khususnya ketika aspek atau opini tidak tertulis secara eksplisit (Li et al., 2023; Cai et al., 2021; Zhang et al., 2024). Oleh karena itu, ACOSQE tidak bersaing melawan model generatif; model generatif justru menjadi salah satu paradigma utama yang paling banyak digunakan untuk menyelesaikan ACOSQE (Li et al., 2023; Xiong et al., 2023; Peper & Wang, 2022).

Paradigma Model

Penelitian ACOSQE umumnya membandingkan metode generatif berbasis *sequence-to-sequence* dengan metode non-generatif seperti *pipeline* klasifikasi atau penandaan kisi (*grid tagging*) (Li et al., 2023; Peper & Wang, 2022; Zhu et al., 2023). Pendekatan non-generatif bertahap sering mengalami perambatan galat (*error propagation*) antartahap (Li et al., 2023; Xu et al., 2026; Xiong et al., 2023). Sebaliknya, model generatif menyelesaikan ekstraksi kuadruplet secara terpadu (*end-to-end*), memanfaatkan semantik label bahasa alami secara fleksibel (Li et al., 2023; Nie et al., 2024; Hoang et al., 2022).

Model generatif modern sering mengungguli metode ekstraksi diskriminatif tradisional pada tolak ukur ACOS (Li et al., 2023; Xiong et al., 2023; Lei et al., 2024). Pengecualian terjadi pada arsitektur non-generatif khusus yang dirancang dengan dekomposisi pemahaman bacaan mesin (*machine reading comprehension* atau MRC) atau *grid tagging*, yang dapat menandingi atau melampaui beberapa varian generatif standar (Zhu et al., 2023; Zhang et al., 2024).



Fig 1. An example of the aspectcategory-opinion-sentiment quadruple extraction task with implicit aspects and opinions (Zhang et al., 2024)
How to read: Diagram ini mengilustrasikan contoh kalimat ulasan yang memuat elemen aspek dan opini implisit beserta keterhubungannya dalam struktur kuadruplet ACOSQE.

Tantangan Opini Implisit

Opini implisit (*implicit opinion*) menjadi bagian paling menantang dalam dataset ACOSQE karena tidak memiliki rentang kata nyata di dalam teks (Cai et al., 2021; Peper & Wang, 2022; Wang et al., 2024). Pada dataset ACOS standar, elemen implisit dipetakan ke token penanda khusus seperti `NULL` (Cai et al., 2021; Xu et al., 2023). Kalimat dengan opini implisit kerap mencatat skor performa terendah dibandingkan kalimat yang seluruh elemennya tersurat (Cai et al., 2021; Peper & Wang, 2022).

Untuk mengatasi kelemahan model generatif dasar dalam menangani kasus implisit, berbagai teknik augmentasi representasi telah dikembangkan (Peper & Wang, 2022; Wang et al., 2024; Lei et al., 2024). Penambahan *supervised contrastive learning* (SCL) mendorong representasi terpisah antara kalimat eksplisit dan implisit (Peper & Wang, 2022; Lei et al., 2024). Model generatif non-autoregresif dengan variabel laten juga dikembangkan untuk mengoptimalkan penangkapan struktur emosi laten tanpa penurunan efisiensi (Nie et al., 2024).

Perbandingan Arsitektur

Model	Paradigma	Penanganan Implisit	Peningkatan Utama	Citations
Extract-Classify	Pipeline Non-generatif	Token khusus klasifikasi	Baseline terstruktur awal	(Cai et al., 2021; Peper & Wang, 2022)

Model	Paradigma	Penanganan Implisit	Peningkatan Utama	Citations
SG-ACOS	Generatif T5	Khusus token peringkas	Linearisasi target pendek	(Li et al., 2023)
GEN-SCL-NAT	Generatif autoregresif	*Contrastive learning*	Target berurutan kausal	(Peper & Wang, 2022)
DPML	Generatif terpandu *prompt*	Pembelajaran timbal-balik	Interaksi *dual-template*	(Lei et al., 2024)
MRC-CLRI	Non-generatif multi-tugas	Dekomposisi relasi query	Inferensi dua arah teroptimasi	(Zhang et al., 2024)

FIGURE 2 Perbandingan model penyelesai tugas ACOSQE pada ulasan implisit

	REST	LAPTOP	LAPTOP-L1
#Categories	13	121	21
#Sentences	2286	4076	4076
#EAEO Quads	2429 (66.4%)	3269 (56.8%)	3269 (56.8%)
#IAEO Quads	530 (14.5%)	910 (15.8%)	910 (15.8%)
#EAIO Quads	350 (9.57%)	1237 (21.5%)	1237 (21.5%)
#IAIO Quads	349 (9.54%)	342 (5.94%)	342 (5.94%)
#Quads/Sent	1.60	1.42	1.42

Table 1. Dataset statistics. Over 33% of REST quadruples contain implicit language, as do 43% in LAPTOP*. E: explicit, I: implicit, A: aspect, O: opinion. E.g., IAEO refers to 'implicit aspect, explicit opinion'. (Peper & Wang, 2022)
How to read: Tabel ini merangkum statistik subset data ulasan restoran dan laptop, mengelompokkan proporsi kuadruplet berdasarkan aspek dan opini eksplisit versus implisit.

Arah Perkembangan Ekstraksi

- Berbagai model generatif menunjukkan **keunggulan konsisten** dibanding metode *pipeline* klasik pada subset implisit (Li et al., 2023; Lei et al., 2024; Wang et al., 2023).
- Penggunaan contoh negatif adaptif (*adaptive negative sampling*) secara signifikan **meningkatkan F1 score** ekstraksi kuadruplet (Xu et al., 2023).
- Model bahasa besar tanpa penyesuaian khusus **mengalami kesulitan evaluasi** akibat ketatnya metrik pencocokan tepat (*exact match*) (Zhang et al., 2024).
- Formulasi multi-tugas berbantuan *prompt* **memperbaiki adaptabilitas lintas bahasa** dan domain spesifik (Xu et al., 2025; Jia et al., 2026; Van Dang et al., 2024).

Secara ringkas, ACOSQE tidak mengungguli model generatif karena ACOSQE merupakan nama tugasnya, sementara model generatif merupakan teknik yang terbukti paling kompetitif dalam menyelesaikan ekstraksi opini implisit tersebut (Li et al., 2023; Peper & Wang, 2022; Lei et al., 2024).

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at https://consensus.app. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

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